

Lecture 8 Team Projects

Embedded Options and Option-Adjusted Measures

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1 Project 1: Callable Bond Relative-Value Review

Scenario. Choose between a callable corporate bond and a comparable option-free alternative.

Objectives. Compare yield to worst, static spread, call schedule, and reinvestment risk; explain the embedded call; evaluate rate-volatility scenarios.

Sources. FINRA TRACE; SEC EDGAR or SEDAR+; U.S. Treasury, FRED, or Bank of Canada curves; instructor OAS snapshot.

Deliverables. Term sheet, call timeline, spread comparison, three scenarios, recommendation, and a 10–12 minute presentation.

Rubric (100 points). Terms 20; spread analysis 25; optionality 30; recommendation 15; presentation and sourcing 10.

2 Project 2: Backward Induction in an Interest-Rate Tree

Scenario. Explain audibly how callability or putability changes bond value.

Objectives. Price option-free and embedded-option bonds in a small tree; apply exercise rules; test volatility, exercise-price, and spread sensitivity.

Sources. Instructor rate tree and terms; U.S. Treasury or Bank of Canada curve; realistic issuer exercise schedule.

Deliverables. Node spreadsheet, values, exercise map, three sensitivities, audit note, and a 10–12 minute presentation.

Rubric (100 points). Setup 20; induction 30; exercise logic 20; sensitivities 20; presentation 10.

3 Project 3: Mortgage Extension Risk Case

Scenario. Advise whether an agency mortgage fund suits an investor worried about both falling and rising rates.

Objectives. Explain prepayment, contraction, extension, and negative convexity; compare the fund with a Treasury benchmark; analyze three rate-path scenarios.

Sources. Agency MBS ETF fact sheets; Fannie Mae or Freddie Mac disclosures; FRED mortgage and Treasury rates; instructor analytics.

Deliverables. Cash-flow explanation, benchmark comparison, three scenarios, suitability conclusion, and a 10–12 minute presentation.

Rubric (100 points). Security understanding 25; benchmark analysis 20; scenario reasoning 25; suitability judgment 20; presentation and citations 10.

4 Project 4: OAS Assumption Audit

Scenario. Two vendors report different OAS values for the same callable bond; identify why and decide which is decision-useful.

Objectives. Compare curves, volatility, prepayment/exercise assumptions, and model structure; reconcile static spread and OAS; assess model risk.

Sources. Issuer prospectus; FINRA data; public vendor methodology notes; instructor-provided analytics from two models.

Deliverables. Assumption inventory, OAS reconciliation, sensitivity table, model-risk recommendation, and a 10–12 minute presentation.

Rubric (100 points). Assumption audit 25; reconciliation 25; sensitivities 20; model-risk judgment 20; presentation and attribution 10.